

Late-time slope diagnostics from saved slab-only purification histories

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We analyze existing 100-trajectory slab-only ensembles at seven circumferences through four times the circumference. Paired endpoint differences and late-window linear fits give finite-time size exponents close to one. The analysis removes a constant raw-gap offset under an affine-growth assumption; it does not establish an infinite-time Lyapunov gap. No new dynamics are simulated, and no manuscript gap claims are changed.

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I. ESTIMATORS AND UNCERTAINTY

For each record, the saved raw modular gap is the minimum absolute modular energy, with pinned occupations excluded using the original extraction tolerance 10^{-9} :

$$g_{\text{mod}}(t) = \min_j \left| \log \frac{1 - \nu_j(t)}{\nu_j(t)} \right| = 2t\Delta(t). \quad (1)$$

The independent sampling unit is a trajectory. For each of the 100 trajectories at each circumference, compute

$$\Delta_{\text{late}} = \frac{g_{\text{mod}}(4N_y) - g_{\text{mod}}(3N_y)}{2N_y}. \quad (2)$$

The primary estimate is its ensemble mean, with ordinary trajectory SEM. This retains temporal correlations automatically. A comparison estimator is half the unweighted least-squares slope of g_{mod} on all saved cycles in $[3N_y, 4N_y]$, with a free intercept, computed separately per trajectory and then averaged with its SEM. Endpoint differences and multi-time fits are different estimators. No bootstrap is used. All data use $N_x = 20$, $\alpha_1 = 1$, $\alpha_2 = 30$, and the saved slab-only protocol with a maximally mixed slab and Born-prepared exterior. Full-measurement histories are not pooled. Source receipts, trajectory IDs, and the relation $g_{\text{mod}} = 2t\Delta$ were verified in the existing extraction; this analysis verifies the sample-history checksum and complete paired records.

II. SIZE COMPARISON

TABLE I. Ensemble means with one trajectory SEM. All entries are finite-window rates.

N_y	$\overline{\Delta}(2N_y)$	Endpoint slope	Fitted slope
20	0.04136 ± 0.00248	0.06688 ± 0.00629	0.06534 ± 0.00653
24	0.03736 ± 0.00185	0.05912 ± 0.00468	0.05767 ± 0.00460
30	0.02803 ± 0.00146	0.04395 ± 0.00364	0.04365 ± 0.00383
36	0.02234 ± 0.00141	0.03459 ± 0.00310	0.03354 ± 0.00322
44	0.01790 ± 0.00091	0.02872 ± 0.00249	0.02770 ± 0.00270
56	0.01467 ± 0.00080	0.02186 ± 0.00194	0.02200 ± 0.00214
60	0.01317 ± 0.00072	0.02255 ± 0.00194	0.02158 ± 0.00206

The fits use $\log \Delta = \log A - z \log N_y$, with propagated log-space SEMs $\sigma_{\log \Delta} = \text{SEM}/\Delta$. Exponent errors are the formal weighted-regression covariance errors from these supplied uncertainties, without rescaling by residual scatter.

$$\begin{aligned} z_{2N_y} &= 1.0788 \pm 0.0540, & \chi^2/\text{dof} &= 0.489, \\ z_{\text{endpoint}} &= 1.0682 \pm 0.0859, & \chi^2/\text{dof} &= 0.332, \\ z_{\text{fit}} &= 1.0751 \pm 0.0921, & \chi^2/\text{dof} &= 0.238. \end{aligned} \tag{3}$$

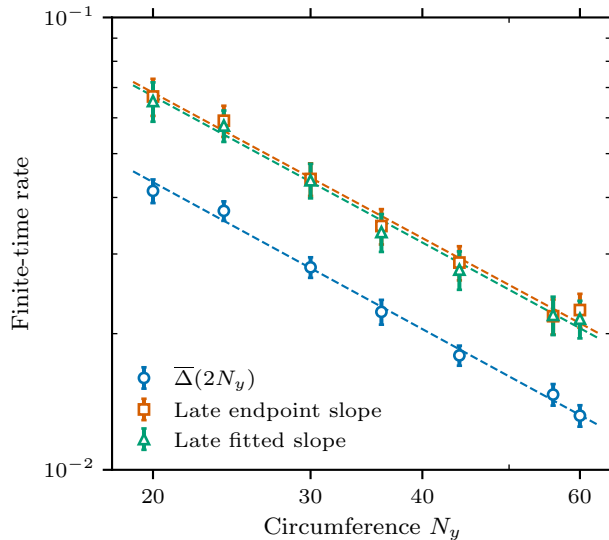


FIG. 1. Saved slab-only finite-time rates versus circumference. Bars are paired trajectory SEMs; dashed curves are separate SEM-weighted power-law fits. All 100 trajectories are retained at each size.

III. INTERPRETATION AND LIMITS

Both late-window estimators support an approximately inverse-circumference trend over the seven saved sizes, rather than an evident size-independent plateau. Under $g_{\text{mod}}(t) \simeq at + b$, a slope removes the constant offset b and yields $a/2$, whereas $\Delta(t)$ contains $b/(2t)$. This cancellation is conditional on approximately affine growth over the selected window. The two slope estimates differ by less than 1.4 paired SEM at every size. The late estimates exceed $\overline{\Delta}(2N_y)$, consistent with the previously observed drift. Their agreement in size exponent strengthens the finite-time evidence: the observed trend is not removed by subtracting a constant raw-gap offset. However, the minimizing mode can change with cycle, and the observation windows themselves grow with N_y . These are not established asymptotic exponents or infinite-time gaps. No new results have been inserted into Figures 4 or A2, or used to change the section title or gap claims.